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Battery cell, battery, and electrical device

Abstract

A battery cell includes a shell, an electrode assembly, and a first insulating member. The shell has an opening and a bottom wall opposite to the opening. The electrode assembly is accommodated in the shell. The first insulating member is used for separating the electrode assembly and the bottom wall. The first insulating member is provided with a first through hole, and projections of the electrode assembly and the first through hole on the bottom wall partially overlap or do not overlap in a thickness direction of the bottom wall.

Inventors: Bai; Lulu (Ningde, CN), Chen; Xinxiang (Ningde, CN), Wu; Ningsheng (Ningde, CN), Zheng; Ting (Ningde, CN), Zhou; Wenlin (Ningde, CN), Zheng; Yulian (Ningde, CN), Li; Quankun (Ningde, CN)

Applicant: CONTEMPORARY AMPEREX TECHNOLOGY (HONG KONG) LIMITED
(Hong Kong, CN)

Family ID: 88328710

Assignee: CONTEMPORARY AMPEREX TECHNOLOGY (HONG KONG) LIMITED
(Hong Kong, CN)

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Primary Examiner: Rhee; Jane J

Attorney, Agent or Firm: ANOVA LAW GROUP, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of International Application No. PCT/CN2022/086390, filed on Apr. 12, 2022, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(1) The present application relates to the field of battery technologies, and in particular, to a battery cell, a battery, and an electrical device.

BACKGROUND ART

(2) At present, with the rapid development of smart phones, tablet computers, electric vehicles, and the like, the application of lithium-ion batteries is increasingly wide, which has put forward higher requirements for lithium-ion batteries. For example, batteries are required to have better safety performance. Internal short circuits in batteries are one of the main reasons for power safety problems.

(3) The short circuits in batteries may generate excessive electric heat and high temperatures, cause fires, and burn out electrical appliances, to threaten property and life safety. Therefore, how to reduce the risk of short circuits in batteries has become an urgent problem in the field of battery technologies.

SUMMARY

(4) Embodiments of the present application provide a battery cell, a battery, and an electrical device to reduce the risk of a short circuit inside the battery cell.

(5) In a first aspect, an embodiment of the present application provides a battery cell, including a shell, an electrode assembly, and a first insulating member; the shell has an opening and a bottom wall opposite to the opening; the electrode assembly is accommodated in the shell; the first insulating member is used for separating the electrode assembly and the bottom wall; and the first insulating member is provided with a first through hole, and projections of the electrode assembly and the first through hole on the bottom wall partially overlap or do not overlap in a thickness direction of the bottom wall.

(6) In the foregoing technical solution, the projections of the electrode assembly and the first through hole on the bottom wall partially overlap or do not overlap in the thickness direction of the bottom wall, that is, at least a portion of the projection of the first through hole on the bottom wall is located outside the area covered by the projection of the electrode assembly on the bottom wall in the thickness direction of the bottom wall, the electrode assembly and the first through hole are misaligned, and there is a very small channel or no channel extending in the thickness direction of the bottom wall between the electrode assembly and the bottom wall of the shell, thereby reducing the risk that ions of the electrode assembly can reach the shell via the first through hole, and also reducing the risk that the ions of the electrode assembly reach the shell to cause a short circuit in the battery cell. Moreover, as the risk that the ions of the electrode assembly reach the shell is reduced, the risk of corrosion of the shell after the ions are in contact with the shell can also be reduced.

(7) In some embodiments of the first aspect of the present application, the electrode assembly has a first end face facing the bottom wall, and the first end face exposes electrode plates of the electrode assembly.

(8) In the foregoing technical solution, the first end face exposes the electrode plates of the electrode assembly, and ions of the electrode plates can deintercalate from the first end face. As the first end face faces the bottom wall and is misaligned with the first through hole on the first insulating member, there is a very small channel or no channel extending in the thickness direction of the bottom wall between the electrode assembly and the bottom wall of the shell, thereby reducing the risk that the ions of the electrode assembly can reach the shell via the first through hole, and also reducing the risk that the ions of the electrode assembly reach the shell to cause a short circuit in the battery cell. Moreover, as the risk that the ions of the electrode assembly reach the shell is reduced, the risk of corrosion of the shell after the ions are in contact with the shell can also be reduced.

(9) In some embodiments of the first aspect of the present application, the electrode assembly

includes a first electrode plate and a second electrode plate with opposite polarities, and the first electrode plate and the second electrode plate are wound around a winding axis perpendicular to the bottom wall; or the electrode assembly includes a separator, a plurality of first electrode plates, and a plurality of second electrode plates, where the first electrode plates and the second electrode plates have opposite polarities, the plurality of first electrode plates and the plurality of second electrode plates are alternately laminated in a first direction, the separator is continuously bent and includes a plurality of first laminated segments and a plurality of first bent segments, each of the first bent segments is used for connecting two adjacent first laminated segments, the plurality of first bent segments are distributed at two ends of the electrode assembly in a second direction, and the first direction, the second direction, and the thickness direction of the bottom wall are perpendicular to each other; or in some embodiments of the first aspect of the present application, the electrode assembly includes a plurality of first electrode plates and a second electrode plate, the first electrode plates and the second electrode plate have opposite polarities, the second electrode plate is continuously bent and includes a plurality of second laminated segments and a plurality of second bent segments, the plurality of second laminated segments and the plurality of first electrode plates are alternately laminated in the first direction, each of the second bent segments is used for connecting two adjacent second laminated segments, the plurality of second bent segments are distributed at two ends of the electrode assembly in the second direction, and the first direction, the second direction, and the thickness direction of the bottom wall are perpendicular to each other.

(10) In the foregoing technical solution, the first electrode plate and second electrode plate of the electrode assembly are wound around the winding axis to form a wound electrode assembly, one end of the electrode assembly along the winding axis exposes the electrode plates, and the ions of the electrode plate can deintercalate from an end, facing the bottom wall, of the electrode assembly along the winding axis. The end, exposing the electrode plates, of the laminated electrode assembly faces the bottom wall, the electrode assembly is misaligned with the first through hole on the first insulating member, and there is a very small channel or no channel extending in the thickness direction of the bottom wall between the electrode assembly and the bottom wall of the shell, thereby reducing the risk that the ions of the electrode assembly can reach the shell via the first through hole, and also reducing the risk that the ions of the electrode assembly reach the shell to cause a short circuit in the battery cell. Moreover, as the risk that the ions of the electrode assembly reach the shell is reduced, the risk of corrosion of the shell after the ions are in contact with the shell can also be reduced.

(11) In some embodiments of the first aspect of the present application, the shell has side walls, and the side walls are disposed on a periphery of the bottom wall; and the electrode assembly is a wound electrode assembly and includes a straight portion and two bent portions, the two bent portions are connected to two ends of the straight portion separately, and the first through hole is disposed between the bent portion and the side wall.

(12) In the foregoing technical solution, the first through hole is disposed between the bent portion and the side wall to facilitate the arrangement of the first through hole.

(13) In some embodiments of the first aspect of the present application, the first insulating member is rectangular, and the first through hole is disposed in at least one of four corners of the first insulating member.

(14) In the foregoing technical solution, the first through hole is disposed in at least one of the four corners of the first insulating member, which facilitates the arrangement of the first through hole, facilitates positioning with an assembly apparatus via the first through hole during assembly of the battery cell, and avoids interference between the electrode assembly and the assembly apparatus during the assembly.

(15) In some embodiments of the first aspect of the present application, the battery cell includes a plurality of electrode assemblies arranged side by side; and in an arrangement direction of the plurality of electrode assemblies, the first through hole is disposed between two adjacent bent

portions.

(16) In the foregoing technical solution, the first through hole is disposed between the two adjacent bent portions to facilitate the arrangement of the first through hole.

(17) In some embodiments of the first aspect of the present application, two groups of first through holes are disposed between two adjacent electrode assemblies, the two groups of first through holes are spaced, and some straight portions are located between the two groups of first through holes.

(18) In the foregoing technical solution, some straight portions are located between the two groups of first through holes, that is, the first through holes are disposed at two ends of the straight portions where the bent portions are disposed, and the straight portions of the two adjacent electrode assemblies can be closely disposed without reserving any space to avoid the first through holes, thereby fully using the internal space of the shell and increasing the energy density of the battery cell. In addition, some straight portions are located between the two groups of first through holes, so that the through holes are distributed more uniformly. During the assembly of the battery cell, if the first insulating member is positioned by the first through holes, the stability and accuracy of positioning can be improved, and the assembly quality of the battery cell can also be improved.

(19) In some embodiments of the first aspect of the present application, the shell further includes side walls, and the side walls are disposed on the periphery of the bottom wall; and the battery cell further includes a second insulating member, and the second insulating member is used for separating the electrode assembly and the side walls.

(20) In the foregoing technical solution, the second insulating member separates the electrode assembly and the side walls to prevent the contact between the electrode assembly and the side walls from causing a short circuit in the battery cell, so as to improve the safety of the battery cell.

(21) In some embodiments of the first aspect of the present application, the second insulating member is integrally formed with the first insulating member.

(22) In the foregoing technical solution, the second insulating member is integrally formed with the first insulating member, which can better separate the bottom wall of the shell and the electrode assembly, as well as the side walls of the shell and the electrode assembly, to avoid a short circuit in the battery cell, so as to improve the safety performance of the battery cell.

(23) In some embodiments of the first aspect of the application, the battery cell further includes a support member, the support member is disposed between the first insulating member and the bottom wall, and the support member is provided with a second through hole corresponding to the first through hole.

(24) In the foregoing technical solution, the support member is disposed between the first insulating member and the bottom wall, which can raise the position of the electrode assembly relative to the bottom wall, and reduce the risk of wrinkling of the electrode plates due to interference between the connection positions of the side walls and bottom wall of the shell and the electrode plates, thereby reducing the risk of a short circuit in the battery cell. The support member may be positioned with the assembly apparatus for assembling the battery cell via the second through hole, thereby improving the assembly quality. The support member is provided with the second through hole corresponding to the first through hole to facilitate accurate positioning between the first insulating member and the support member, so as to improve the assembly quality.

(25) In a second aspect, an embodiment of the present application provides a battery, including the battery cell provided by the embodiments in the first aspect.

(26) In the foregoing technical solution, there is no channel extending in the thickness direction of the bottom wall between the electrode assembly of the battery cell and the bottom wall of the shell, thereby eliminating a linear ion channel between the electrode assembly and the bottom wall, reducing the risk that ions of the electrode assembly can reach the shell via the first through hole, reducing the risk that the ions of the electrode assembly reach the shell to cause a short circuit in

the battery cell, and improving the safety performance of the battery accordingly.

(27) In a third aspect, an embodiment of the present application provides an electrical device, including the battery cell provided by the embodiments in the first aspect.

(28) In the foregoing technical solution, there is no channel extending in the thickness direction of the bottom wall between the electrode assembly of the battery cell and the bottom wall of the shell, thereby eliminating a linear ion channel between the electrode assembly and the bottom wall, reducing the risk that ions of the electrode assembly can reach the shell via the first through hole, reducing the risk that the ions of the electrode assembly reach the shell to cause a short circuit in the battery cell, and improving the power safety of the electrical device accordingly.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) In order to explain the technical solutions in the embodiments of the present application more clearly, accompanying drawings required to use in the embodiments will be simply introduced below. It should be understood that the following drawings show only some embodiments of the present application and should not be regarded as limiting the scope, and other relevant drawings may be obtained based on these drawings by those of ordinary skill in the art without any creative efforts.

(2) FIG. 1 is a schematic structural diagram of a vehicle provided by some embodiments of the present application;

(3) FIG. 2 is a schematic structural diagram of a battery provided by some embodiments of the present application;

(4) FIG. 3 is an exploded view of a battery cell provided by some embodiments of the present application;

(5) FIG. 4 is a schematic diagram of a relative relationship between a first insulating member and an electrode assembly provided by some embodiments of the present application;

(6) FIG. 5 is a schematic structural diagram of an electrode assembly provided by some embodiments of the present application;

(7) FIG. 6 is a schematic structural diagram of an electrode assembly provided by other embodiments of the present application;

(8) FIG. 7 is a schematic structural diagram of an electrode assembly provided by still other embodiments of the present application;

(9) FIG. 8 is a schematic diagram of a relative relationship between a first insulating member and an electrode assembly provided by other embodiments of the present application;

(10) FIG. 9 is a schematic diagram of a relative relationship between a first insulating member and an electrode assembly provided by still other embodiments of the present application;

(11) FIG. 10 is a schematic structural diagram of a shell provided by some embodiments of the present application;

(12) FIG. 11 is a schematic structural diagram of a first insulating member and a second insulating member after connection according to some embodiments of the present application;

(13) FIG. 12 is a schematic structural diagram after first separation portions and second separation portions of the first insulating member are folded according to some embodiments of the present application;

(14) FIG. 13 is a partial cross-sectional view of a battery cell provided by some embodiments of the present application;

(15) FIG. 14 is a schematic structural diagram of a manufacturing device for a battery cell provided by some embodiments of the present application; and

(16) FIG. 15 is a flowchart of a manufacturing method for a battery cell provided by some

embodiments of the present application.

(17) Reference numerals: **1000**—vehicle; **100**—battery; **10**—box; **11**—mounting space; **12**—first portion; **13**—second portion; **20**—battery cell; **21**—shell; **211**—opening; **212**—bottom wall; **213**—side wall; **2131**—first side wall; **2132**—second side wall; **22**—electrode assembly; **I**—straight portion; **II**—bent portion; **221**—first electrode plate; **222**—second electrode plate; **2221**—second laminated segment; **2222**—second bent segment; **223**—separator; **2231**—first laminated segment; **2232**—first bent segment; **23**—end cover assembly; **231**—end cover; **232**—electrode terminal; **233**—end cover protector; **24**—protective film; **25**—current collecting member; **26**—first insulating member; **261**—first through hole; **262**—first crease; **27**—second insulating member; **271**—first separation portion; **272**—second separation portion; **273**—second crease; **28**—support member; **281**—second through hole; **200**—controller; **300**—motor; **2000**—manufacturing device for a battery cell; **2100**—provision apparatus; **2200**—assembly apparatus; **X**—thickness direction of the bottom wall; **Y**—first direction; **Z**—second direction; **M**—fourth direction; **N**—Fifth direction.

DETAILED DESCRIPTION OF EMBODIMENTS

(18) In order to make the objectives, technical solutions and advantages of the embodiments of the present application clearer, the technical solutions in the embodiments of the present application will be described clearly and completely below in conjunction with the accompanying drawings in the embodiments of the present application. Apparently, the described embodiments are some of the embodiments of the present application, not all of them. Generally, components of the embodiments of the present application described and shown in the drawings may be arranged and designed with various different configurations.

(19) Therefore, the detailed descriptions of the embodiments of the present application provided in the drawings below are not intended to limit the scope of the present application, but are merely representatives of the selected embodiments of the present application. All other embodiments obtained by those of ordinary skill in the art based on the embodiments of the present application without creative efforts shall fall within the protection scope of the present application.

(20) It should be noted that the embodiments in the present application and the features in the embodiments may be combined with each other on a non-conflict basis.

(21) It should be noted that similar reference numerals and letters in the following drawings represent similar terms, so once a term is defined in a drawing, further definition and explanation are not required in the follow-up drawings.

(22) In the description of the embodiments of the present application, indicative orientations or positional relationships are orientations or positional relationships shown based on the drawings, or commonly placed orientations or positional relationships when the product of the present application is used, or orientations or positional relationships commonly understood by those skilled in the art, are merely for convenience of describing the present application and for simplifying the description, but not for indicating or implying that an indicated apparatus or element must have a specific orientation, and must be constructed and operated in a specific orientation, and therefore, cannot be understood as limiting the present application. In addition, the terms “first”, “second”, “third”, and the like are merely used for distinguishing the description and cannot be construed as indicating or implying relative importance.

(23) At present, in view of the development of the market situation, use of power batteries is increasingly widespread. Power batteries are not only used in energy storage power systems such as hydropower, firepower, wind power, and solar power plants, but also widely used in electric vehicles such as electric bicycles, electric motorcycles, and electric vehicles, as well as in various fields such as military equipment and aerospace. With the continuous expansion of the application fields of power batteries, their market demand is also increasing.

(24) A battery cell includes a shell, an electrode assembly, and a first insulating member. The first insulating member is used for separating the electrode assembly from a bottom wall of the shell, so as to avoid a contact short circuit between the electrode assembly and the bottom wall of the shell.

In order to facilitate the assembly of the battery cell, the first insulating member is provided with a through hole. The through hole is used for fitting an assembly apparatus for assembling the battery cell to position the first insulating member on the assembly apparatus, so as to improve assembly quality.

(25) The inventor found that ions in the electrode assembly reach the shell via the through hole to result in a short circuit inside the battery cell and cause safety problems. To prevent ions from reaching the shell via the through hole, the through hole may be sealed by pasting an adhesive tape on the first insulating member. However, the adhesive tape falls off and fails in sealing when immersed in an electrolytic solution for a long term. Moreover, after the ions of the electrode assembly are in contact with the shell, they may undergo chemical reactions with the shell to corrode the shell.

(26) Based on the above considerations, in order to reduce passing of ions via the through hole, the inventor conducted in-depth research and designed a battery cell, where the battery cell includes a first insulating member used for separating an electrode assembly and a bottom wall of a shell, the first insulating member is provided with a first through hole, and projections of the electrode assembly and the first through hole on the bottom wall do not overlap in a thickness direction of the bottom wall. That is, the electrode assembly and the through hole are misaligned, the electrode assembly will not cover the first through hole, and there is no linear channel for ions to pass through between the electrode assembly and the bottom wall of the shell. Therefore, linear ion channels are eliminated, the risk that ions of the electrode assembly can reach the shell via the first through hole is reduced, and the risk that the ions of the electrode assembly reach the shell to cause a short circuit in the battery cell is also reduced. Moreover, as the risk that the ions of the electrode assembly reach the shell is reduced, the risk of shell corrosion caused by contact between the ions and the shell can also be reduced.

(27) The battery cell disclosed in the embodiments of the present application may be used, but not limited to, in an electrical device such as a vehicle, a ship, or an aircraft. A power system of the electrical device may be constituted by the battery cell, the battery, or the like disclosed in the present application. In this case, the risk of passing of ions via the through hole is reduced, and the risk of a short circuit inside the battery is also reduced, thereby improving the safety performance of the battery cell.

(28) The technical solutions described in the embodiments of the present application are applicable to batteries and electrical devices using batteries.

(29) The electrical device may be a vehicle, a mobile phone, a portable device, a notebook computer, a ship, a spacecraft, an electric toy, an electric tool, or the like. The vehicle may be an oil-fueled vehicle, a gas vehicle, or a new energy vehicle. The new energy vehicle may be a battery electric vehicle, a hybrid electric vehicle, an extended-range vehicle, or the like. The spacecraft includes an airplane, a rocket, a space shuttle, a spaceship, or the like. The electric toy includes a fixed or mobile electric toy, such as a game console, an electric car toy, an electric ship toy, or an electric aircraft toy. The electric tool includes a metal cutting electric tool, a grinding electric tool, an assembly electric tool, or a railway electric tool, such as an electric drill, an electric grinder, an electric wrench, an electric screwdriver, an electric hammer, an impact electric drill, a concrete vibrator, or an electric planer. The embodiments of the present application do not limit the foregoing electrical devices.

(30) For convenient description, the following embodiments are described by an example of a vehicle as an electrical device.

(31) With reference to FIG. 1, a battery **100** is disposed inside a vehicle **1000**, and the battery **100** may be disposed at a bottom, head, or tail of the vehicle **1000**. The battery **100** may be used for supplying power to the vehicle **1000**. For example, the battery **100** may be used as an operation power supply of the vehicle **1000**.

(32) The vehicle **1000** may further include a controller **200** and a motor **300**. The controller **200** is

used for controlling the battery **100** to supply power to the motor **300**, for example, for a working power demand of the vehicle **1000** during startup, navigation and running.

(33) In some embodiments of the present application, the battery **100** may be used not only as an operation power supply of the vehicle **1000**, but also as a driving power supply of the vehicle **1000** to replace or partially replace fuel or natural gas to provide driving power for the vehicle **1000**.

(34) With reference to FIG. **2**, the battery **100** includes a box **10** and battery cells **20**, where the battery cells **20** are accommodated in the box **10**.

(35) The box **10** is used for providing a mounting space **11** for the battery cells **20**. In some embodiments, the box **10** may include a first portion **12** and a second portion **13**, and the first portion **12** and the second portion **13** cover each other to jointly confine the mounting space **11** for accommodating the battery cells **20**. Of course, the connection between the first portion **12** and the second portion **13** may be sealed by a sealing member (not shown), which may be a sealing ring, a sealant, or the like.

(36) The first portion **12** and the second portion **13** may be in various shapes, such as rectangles or cylinders. The first portion **12** may be opened on one side to form a hollow structure with an accommodating cavity for accommodating the battery cells **20**, the second portion **13** may also be opened on one side to form a hollow structure with an accommodating cavity for accommodating the battery cells **20**, and the open side of the second portion **13** covers the open side of the first portion **12** to form the box **10** with the mounting space **11**. Alternatively, the first portion **12** may be opened on one side to form a hollow structure with an accommodating cavity for accommodating the battery cells **20**, the second portion **13** is a of plate-like structure, and the second portion **13** covers the open side of the first portion **12** to form the box **10** with the mounting space **11**.

(37) There may be one or more battery cells **20** in the battery **100**. If there is a plurality of battery cells **20**, the plurality of battery cells **20** may be in series connection, parallel connection, or series-parallel connection. The series-parallel connection refers to a combination of series connection and parallel connection in the plurality of battery cells **20**. The plurality of battery cells **20** may be directly connected in series, in parallel, or in series and parallel together, and then the whole formed by the plurality of battery cells **20** is accommodated in the box **10**. Alternatively, the plurality of battery cells **20** may be first connected in series, in parallel, or in series and parallel to form a battery module, then a plurality of battery modules are connected in series, in parallel, or in series and parallel to form a whole, and the whole is accommodated in the box **10**. The battery cells **20** may be cylindrical, flat, cuboid, or in other shapes. FIG. **2** illustrates square battery cells **20**.

(38) In some embodiments, the battery **100** may further include a current collecting component (not shown), and the plurality of battery cells **20** may be electrically connected through the current collecting component to implement the parallel, series, or series-parallel connection of the plurality of battery cells **20**.

(39) With reference to FIG. **3**, the battery cell **20** may include a shell **21**, an electrode assembly **22**, and an end cover assembly **23**. The shell **21** has an opening **211**, the electrode assembly **22** is accommodated in the shell **21**, and the end cover assembly **23** is used for covering the opening **211**.

(40) The shell **21** may be in various shapes, such as cylindrical or cuboid. The shape of the shell **21** may be determined according to a specific shape of the electrode assembly **22**. For example, if the electrode assembly **22** is of a cylindrical structure, the shell **21** may be of a cylindrical structure; or if the electrode assembly **22** is of a rectangular structure, the shell **21** may be of a rectangular structure. FIG. **3** illustrates that the shell **21** and the electrode assembly **22** are square.

(41) A material of the shell **21** may be various, such as copper, iron, aluminum, stainless steel, or aluminum alloy, and is not specially limited in the embodiments of the present application.

(42) In some embodiments, a protective film **24** is further disposed on an outer surface of the shell **21**, and the protective film **24** covers the outer surface of the shell **21**. The protective film **24** can achieve effects of insulation, high temperature resistance, and the like. The protective film **24** may be a blue film.

(43) The electrode assembly **22** may include a positive electrode plate (not shown), a negative electrode plate (not shown), and a separator **223** (not shown). The electrode assembly **22** may be of a wound structure formed by winding the positive electrode plate, the separator **223**, and the negative electrode plate, or a laminated structure formed by laminating the positive electrode plate, the separator **223**, and the negative electrode plate. The electrode assembly **22** further includes a positive electrode tab (not shown) and a negative electrode tab (not shown), a positive electrode current collector not coated with a positive electrode active material layer in the positive electrode plate may be used as the positive electrode tab, and a negative electrode current collector not coated with a negative electrode active material layer in the negative electrode plate may be used as the negative electrode tab.

(44) The end cover assembly **23** includes an end cover **231** and electrode terminals **232**, where the electrode terminals **232** are disposed on the end cover **231**. The end cover **231** is used for covering the opening **211** of the shell **21** to form a closed accommodating space (not shown), and the accommodating space is used for accommodating the electrode assembly **22**. The accommodating space is further used for accommodating an electrolyte, such as an electrolytic solution. The end cover assembly **23** serves as a component for outputting electrical energy of the electrode assembly **22**. The electrode terminals **232** in the end cover assembly **23** are used for electrical connection with the electrode assembly **22**, that is, the electrode terminals **232** are electrically connected to the tabs of the electrode assembly **22**. For example, the electrode terminals **232** are connected to the tabs through current collecting members **25** to achieve electrical connection between the electrode terminals **232** and the tabs.

(45) It should be noted that the shell **21** may have one or two openings **211**. If the shell **21** has one opening **211**, there may also be one end cover assembly **23**, two electrode terminals **232** may be disposed in the end cover assembly **23**, the two electrode terminals **232** are used for electrical connection with the positive electrode tab and negative electrode tab of the electrode assembly **22** separately, and the two electrode terminals **232** in the end cover assembly **23** are a positive electrode terminal **232** and a negative electrode terminal **232** separately. If the shell **21** has two openings **211**, for example, the two openings **211** are disposed on two opposite sides of the shell **21**, there may also be two end cover assemblies **23**, and the two end cover assemblies **23** cover the two openings **211** of the shell **21**. In this case, the electrode terminal **232** in one end cover assembly **23** is a positive electrode terminal **232** used for electrical connection with the positive electrode tab of the electrode assembly **22**; and the electrode terminal **232** in the other end cover assembly **23** is a negative electrode terminal **232** used for electrical connection with the negative electrode tab of the electrode assembly **22**.

(46) In some embodiments, the end cover assembly **23** further includes an end cover protector **233**, and the end cover protector **233** is mounted on a surface of the end cover **231** to protect the end cover **231**.

(47) As shown in FIG. 4, in some embodiments, the battery cell **20** includes a shell **21**, an electrode assembly **22**, and a first insulating member **26**; the shell **21** has an opening **211** and a bottom wall **212** opposite to the opening **211**; the electrode assembly **22** is accommodated in the shell **21**; the first insulating member **26** is used for separating the electrode assembly **22** from the bottom wall **212**; and the first insulating member **26** is provided with a first through hole **261**, and projections of the electrode assembly **22** and the first through hole **261** on the bottom wall **212** partially overlap or do not overlap in a thickness direction of the bottom wall **212**.

(48) The opening **211** of the shell **21** is opposite to the bottom wall **212** in the thickness direction X of the bottom wall. The electrode assembly **22** enters the shell **21** from the opening **211**. In the thickness direction X of the bottom wall, the first insulating member **26** is located between the electrode assembly **22** and the bottom wall **212**. The first insulating member **26** is used for separating the electrode assembly **22** from the bottom wall **212**, indicating that the first insulating member **26** is located between the electrode assembly **22** and the bottom wall **212** to prevent

contact between the electrode assembly **22** and the bottom wall **212**, so as to avoid a short circuit in the battery cell **20** due to the contact between the electrode assembly **22** and the bottom wall **212**.

(49) The projections of the electrode assembly **22** and the first through hole **261** on the bottom wall **212** partially overlap in the thickness direction X of the bottom wall, indicating that a portion of the projection of the first through hole **261** on the bottom wall **212** is located outside an area covered by the projection of the electrode assembly **22** on the bottom wall **212** in the thickness direction X of the bottom wall, and a channel extending in the thickness direction X of the bottom wall between the electrode assembly **22** and the bottom wall **212** of the shell **21** is very small.

(50) The projections of the electrode assembly **22** and the first through hole **261** on the bottom wall **212** do not overlap in the thickness direction X of the bottom wall, indicating that all of the projection of the first through hole **261** on the bottom wall **212** is located outside an area covered by the projection of the electrode assembly **22** on the bottom wall **212** in the thickness direction X of the bottom wall, and there is no channel extending in the thickness direction X of the bottom wall between the electrode assembly **22** and the bottom wall **212** of the shell **21**, that is, a linear ion channel between the electrode assembly **22** and the bottom wall **212** is eliminated.

(51) The projections of the electrode assembly **22** and the first through hole **261** on the bottom wall **212** partially overlap or do not overlap in the thickness direction X of the bottom wall, that is, at least a portion of the projection of the first through hole **261** on the bottom wall **212** is located outside the area covered by the projection of the electrode assembly **22** on the bottom wall **212** in the thickness direction X of the bottom wall, the electrode assembly **22** and the first through hole **261** are misaligned, and there is a very small channel or no channel extending in the thickness direction X of the bottom wall between the electrode assembly **22** and the bottom wall **212** of the shell **21**, thereby reducing the risk that ions of the electrode assembly **22** can reach the shell **21** via the first through hole **261**, and also reducing the risk that the ions of the electrode assembly **22** reach the shell **21** to cause a short circuit in the battery cell **20**. Moreover, as the risk that the ions of the electrode assembly **22** reach the shell **21** is reduced, the risk of corrosion of the shell **21** due to contact between the ions and the shell **21** can also be reduced. When the battery cell **20** is assembled, the first through hole **261** is used for fitting an assembly apparatus **2200** of the battery cell **20** to position the first insulating member **26** on the assembly apparatus **2200**, so as to improve assembly quality. Because the projections of the electrode assembly **22** and the first through hole **261** on the bottom wall **212** do not overlap in the thickness direction X of the bottom wall, the electrode assembly **22** and the assembly apparatus **2200** cannot interfere with each other when the battery cell **20** is assembled.

(52) With reference to FIGS. 5, 6, and 7, in some embodiments, the electrode assembly **22** has a first end face (not shown) facing the bottom wall **212**, and the first end face exposes the electrode plates of the electrode assembly **22**.

(53) The separator **223** of the electrode assembly **22** is used for separating the positive and negative electrode plates, so as to avoid a short circuit in the electrode assembly **22** due to contact between the positive and negative electrode plates. The first end face of the electrode assembly **22** exposes the electrode plates, indicating that the electrode plates are not covered by the separator, and the ions of the electrode assembly **22** can move from the first end face towards the bottom wall **212**. The electrode tabs of the electrode assembly **22** are located on a second end face opposite to the first end face. An end, where the electrode tabs are disposed, of the electrode assembly **22** is not covered by the separator **223**. The tabs of the electrode assembly **22** are generally disposed towards the opening **211** of the shell **21**, the first end face of the electrode assembly **22** faces the bottom wall **212**, and the tabs are located on the second end face opposite to the first end face, thereby facilitating the arrangement of the tabs of the electrode assembly **22** towards the opening **211** of the shell **21**. It can be understood that, for the electrode assembly **22** with a tab on a single side, an end, opposite to the tab, of the electrode assembly **22** is also not covered by the separator **223**. An end, at which the tabs are not disposed and which is not covered by the separator **223**, of the electrode

assembly **22** faces the bottom wall **212**, that is, the first end face exposes the electrode plates of the electrode assembly **22**, and ions of the electrode plates can deintercalate from the first end face. As the first end face faces the bottom wall **212** and is misaligned with the first through hole **261** on the first insulating member **26**, there is a very small channel or no channel extending in the thickness direction X of the bottom wall between the electrode assembly **22** and the bottom wall **212** of the shell **21**, thereby reducing the risk that the ions of the electrode assembly **22** can reach the shell **21** via the first through hole **261**, and also reducing the risk that the ions of the electrode assembly **22** reach the shell **21** to cause a short circuit in the battery cell **20**. Moreover, as the risk that the ions of the electrode assembly **22** reach the shell **21** is reduced, the risk of corrosion of the shell **21** after the ions are in contact with the shell **21** can also be reduced.

(54) The electrode assembly **22** may be in various forms, as shown in FIG. 5, in some embodiments, the electrode assembly **22** includes a first electrode plate **221** and a second electrode plate **222** with opposite polarities, and the first electrode plate **221** and the second electrode plate **222** are wound around a winding axis perpendicular to the bottom wall **212**. The separator **223** is used for separating the first electrode plate **221** and the second electrode plate **222**.

(55) The first electrode plate **221** and the second electrode plate **222** are wound around the winding axis to form a wound electrode assembly. For the wound electrode assembly, two axial ends of the electrode assembly **22** are not covered by the separator **223**, so that both the first electrode plate **221** and the second electrode plate **222** are exposed along the two axial ends of the electrode assembly **22**. All the tabs of the electrode assembly **22** are located at one end (second end face) facing the opening **211** in the axial direction of the electrode assembly **22**, and the other end (first end face) in the axial direction of the electrode assembly **22** faces the bottom wall **212**, so that the winding axis of the electrode assembly **22** is perpendicular to the bottom wall **212** (an extension direction of the winding axis is consistent with the thickness direction X of the bottom wall) to facilitate the connection between the tabs and the electrode terminals **232** on the end cover. The ions of the electrode assembly **22** can move close to the bottom wall **212** in the extension direction of the winding axis. The projection of the electrode assembly **22** on the bottom wall **212** does not overlap the projection of the first through hole **261** on the bottom wall **212** in the thickness direction X of the bottom wall, and there is no channel extending in the thickness direction X of the bottom wall between the electrode assembly **22** and the bottom wall **212** of the shell **21**, thereby eliminating a linear ion channel between the electrode assembly **22** and the bottom wall **212**, reducing the risk that the ions of the electrode assembly **22** can reach the shell **21** via the first through hole **261**, and also reducing the risk that the ions of the electrode assembly **22** reach the shell **21** to cause a short circuit in the battery cell **20**. Moreover, as the risk that the ions of the electrode assembly **22** reach the shell **21** is reduced, the risk of corrosion of the shell **21** after the ions are in contact with the shell **21** can also be reduced.

(56) As shown in FIG. 6, in other embodiments, the electrode assembly **22** includes a separator **223**, a plurality of first electrode plates **221**, and a plurality of second electrode plates **222**, where the first electrode plates **221** and the second electrode plates **222** have opposite polarities, the plurality of first electrode plates **221** and the plurality of second electrode plates **222** are alternately laminated in a first direction Y, the separator **223** is continuously bent and includes a plurality of first laminated segments **2231** and a plurality of first bent segments **2232**, each of the first bent segments **2232** is used for connecting two adjacent first laminated segments **2231**, the plurality of first bent segments **2232** are distributed at two ends of the electrode assembly **22** in a second direction Z, and the first direction Y, the second direction Z, and the thickness direction X of the bottom wall are perpendicular to each other.

(57) In this embodiment, the plurality of first electrode plates **221** are independent of each other, and the plurality of second electrode plates **222** are independent of each other. The separator **223** is of a continuous strip structure. In the first direction Y, a second electrode plate **222** is disposed between two adjacent first electrode plates **221**, and a first electrode plate **221** is disposed between

two adjacent second electrode plates **222**, so that the plurality of first electrode plates **221** and the plurality of second electrode plates **222** are alternately laminated in the first direction Y. In the first direction Y, a first laminated segment **2231** is disposed between the first electrode plate **221** and the second electrode plate **222** that are adjacent, and the two adjacent first laminated segments **2231** are connected by a first bent segment. Some of the plurality of first bent segments **2232** are distributed at one end of the electrode assembly **22** in the second direction Z, and the other first bent segments **2232** are distributed at the other end of the electrode assembly **22** in the second direction Z. The plurality of first bent segments **2232** cover two ends of the first electrode plates **221** and the second electrode plates **222** in the second direction Z. The two ends of the first electrode plates **221** and the second electrode plates **222** in a third direction are not covered by the separator **223**, so that the two ends of the first electrode plates **221** and the second electrode plates **222** are exposed in the third direction. The end, where the tabs are not disposed, of the electrode assembly **22** in the third direction faces the bottom wall **212**. The first direction Y, the second direction Z, and the third direction are perpendicular to each other, and the third direction is consistent with the thickness direction X of the bottom wall. Two ends of the electrode assembly **22** in the thickness direction X of the bottom wall are not covered by the separator **223**, the tabs are located at the end, not covered by the separator **223**, of the electrode assembly **22**, and a direction in which the tabs are located at the two ends, not covered by the separator **223**, of the electrode assembly **22** is consistent with the thickness direction X of the bottom wall and perpendicular to the first direction Y and the second direction Z, thereby facilitating the arrangement of the tabs towards the opening **211** of the shell **21**.

(58) Because the two ends of the first electrode plates **221** and the second electrode plates **222** in the thickness direction X of the bottom wall are not covered by the separator **223**, the ions of the electrode assembly **22** can move close to the bottom wall **212** in the thickness direction X of the bottom wall. The projection of the electrode assembly **22** on the bottom wall **212** does not overlap the projection of the first through hole **261** on the bottom wall **212** in the thickness direction X of the bottom wall, and there is no channel extending in the thickness direction X of the bottom wall between the electrode assembly **22** and the bottom wall **212** of the shell **21**, thereby eliminating a linear ion channel between the electrode assembly **22** and the bottom wall **212**, reducing the risk that the ions of the electrode assembly **22** can reach the shell **21** via the first through hole **261**, and also reducing the risk that the ions of the electrode assembly **22** reach the shell **21** to cause a short circuit in the battery cell **20**. Moreover, as the risk that the ions of the electrode assembly **22** reach the shell **21** is reduced, the risk of corrosion of the shell **21** after the ions are in contact with the shell **21** can also be reduced.

(59) As shown in FIG. 7, in still other embodiments, the electrode assembly **22** includes a plurality of first electrode plates **221** and a second electrode plate **222**, the first electrode plates **221** and the second electrode plate **222** have opposite polarities, the second electrode plate **222** is continuously bent and includes a plurality of second laminated segments **2221** and a plurality of second bent segments **2222**, the plurality of second laminated segments **2221** and the plurality of first electrode plates **221** are alternately laminated in the first direction Y, each second bent segment **2222** is used for connecting two adjacent second laminated segments **2221**, the plurality of second bent segments **2222** are distributed at two ends of the electrode assembly **22** in the second direction Z, and the first direction Y, the second direction Z, and the thickness direction X of the bottom wall are perpendicular to each other.

(60) In this embodiment, the plurality of first electrode plates **221** are independent of each other, and the second electrode plate **222** and the separator **223** are of continuous strip structures. The plurality of second laminated segments **2221** are laminated in the first direction Y, and the two adjacent second laminated segments **2221** are connected by a second bent segment **2222**. In the first direction Y, a second laminated segment **2221** is disposed between two adjacent first electrode plates **221**, and a first electrode plate **221** is disposed between two adjacent second laminated segments **2221**, so that the plurality of second laminated segments **2221** and the plurality of first

electrode plates **221** are alternately laminated in the first direction Y.

(61) The separator **223** separates the second electrode plate **222** and the plurality of first electrode plates **221**, the separator **223** is continuously bent and includes a plurality of first laminated segments **2231** and a plurality of first bent segments **2232**, a first laminated segment **2231** is disposed between the first laminated segment **2231** and the first electrode plate **221** that are adjacent, the two adjacent first laminated segments **2231** are connected by a first bent segment **2232**, and the first bent segments **2232** cover outer sides of the second bent segments **2222**. The plurality of first bent segments **2232** cover two ends of the first electrode plates **221** and the second electrode plate **222** in the second direction Z. The two ends of the first electrode plates **221** and the second electrode plate **222** in a third direction are not covered by the separator **223**, so that the two ends of the first electrode plates **221** and the second electrode plate **222** are exposed in the third direction. The end, where the tabs are not disposed, of the electrode assembly **22** in the third direction faces the bottom wall **212**. The first direction Y, the second direction Z, and the third direction are perpendicular to each other, and the third direction is consistent with the thickness direction X of the bottom wall. Two ends of the electrode assembly **22** in the thickness direction X of the bottom wall are not covered by the separator **223**, the tabs are located at the end, not covered by the separator **223**, of the electrode assembly **22**, and a direction in which the tabs are located at the two ends, not covered by the separator **223**, of the electrode assembly **22** is consistent with the thickness direction X of the bottom wall and perpendicular to the first direction Y and the second direction Z, thereby facilitating the arrangement of the tabs towards the opening **211** of the shell **21**.

(62) With continued reference to FIG. 4, in some embodiments, the shell **21** has side walls **213**, and the side walls **213** are disposed on a periphery of the bottom wall **212**; and the electrode assembly **22** is a wound electrode assembly and includes a straight portion I and two bent portions II, the two bent portions II are connected to two ends of the straight portion I separately, and the first through hole **261** is disposed between the bent portion II and the side wall **213**.

(63) The first through hole **261** is disposed between the bent portion II and the side wall **213**, indicating that the first through hole **261** is disposed in a space enclosed by a peripheral surface of the bent portion II and an inner surface of the side wall **213**. In other embodiments, the first through hole **261** may alternatively be disposed between a straight area and the side wall **213**.

(64) The first through hole **261** is disposed between the bent portion II and the side wall **213** to facilitate the arrangement of the first through hole **261**.

(65) The first insulating member **26** is in various shapes. For example, in some embodiments, as shown in FIG. 4, the first insulating member **26** is rectangular, and the first through hole **261** is disposed in at least one of four corners of the first insulating member **26**.

(66) In an embodiment where the battery cell **20** has one electrode assembly **22**, one or a plurality of first through holes **261** may be disposed on the bottom wall **212**, where the plurality refers to two or more. FIG. 4 shows a case where a plurality of first through holes **261** are disposed on the bottom wall **212**. In FIG. 4, a quantity of the first through holes **261** is two, and the two first through holes **261** are disposed at two diagonals among the four corners of the first insulating member **26** to improve the stability of positioning fit between the first insulating member **26** and the assembly apparatus **2200** via the first through holes **261**.

(67) As shown in FIG. 8, in an embodiment where the battery cell **20** includes a plurality of electrode assemblies **22**, a first through hole **261** may be disposed in at least one of four corners of the first insulating member **26**, and a first through hole **261** may also be disposed between bent portions II of two adjacent electrode assemblies **22**.

(68) In other embodiments, an outer contour of the first insulating member **26** may be in a shape matching an outer contour of the electrode assembly **22**.

(69) The first through hole **261** is disposed in at least one of the four corners of the first insulating member **26**, which facilitates the arrangement of the first through hole **261**, facilitates positioning with an assembly apparatus **2200** via the first through hole **261** during assembly of the battery cell

20, and avoids interference between the electrode assembly **22** and the assembly apparatus **2200** (as shown in FIG. **15**) during the assembly.

(70) As shown in FIG. **8** and FIG. **9**, in some embodiments, the battery cell **20** includes a plurality of electrode assemblies **22** arranged side by side; and in an arrangement direction of the plurality of electrode assemblies **22**, the first through hole **261** is disposed between two adjacent bent portions **II**.

(71) The arrangement direction of the two bent portions **II** of the electrode assemblies **22** is perpendicular to that of the plurality of electrode assemblies **22**.

(72) One first through hole **261** or a plurality of first through holes **261** may be disposed between the two adjacent bent portions **II**. The plurality refers to two or more. Alternatively, a first through hole **261** is disposed between every two adjacent bent portions **II**, or a first through hole **261** is disposed between some of the two adjacent bent portions **II**.

(73) In an embodiment where a first through hole **261** is disposed between two adjacent bent portions **II**, a first through hole **261** may also be disposed in at least one of the four corners of the rectangle of the first insulating member **26** (as shown in FIG. **8**). In other embodiments, a first through hole **261** may be disposed only between two adjacent bent portions **II** (as shown in FIG. **9**).

(74) The first through hole **261** is disposed between the two adjacent bent portions **II** to facilitate the arrangement of the first through hole **261**.

(75) In some embodiments, two groups of first through holes **261** are disposed between two adjacent electrode assemblies **22**, the two groups of first through holes **261** are spaced, and some straight portions **I** are located between the two groups of first through holes **261**.

(76) Each group includes a plurality of first through holes **261**, some first through holes **261** in one group of first through holes **261** located between the two adjacent electrode assemblies **22** are disposed at one end of a straight area, and the other first through holes **261** are disposed at the other end of the straight area, so that some straight portions **I** are located between two first through holes **261**. That is, the first through holes **261** in each group of first through holes **261** are disposed at two ends of the straight portions **I** where the bent portions **II** are disposed, and the straight portions **I** of the two adjacent electrode assemblies **22** can be closely disposed without reserving any space to avoid the first through holes **261**, thereby fully using the internal space of the shell **21** and increasing the energy density of the battery cell **20**. In addition, some straight portions **I** are located between the two first through holes **261**, so that the through holes are distributed more uniformly. During the assembly of the battery cell **20**, if the first insulating member **26** is positioned by the first through holes **261**, the stability and accuracy of positioning can be improved, and the assembly quality of the battery cell **20** can also be improved.

(77) As shown in FIG. **10** and FIG. **11**, in some embodiments, the shell **21** further includes side walls **213**, and the side walls **213** are disposed on a periphery of the bottom wall **212**; and the battery cell **20** further includes a second insulating member **27**, and the second insulating member **27** is used for separating the electrode assembly **22** and the side walls **213**.

(78) The second insulating member **27** may be fixed on a peripheral surface of the electrode assembly **22** or an inner wall of the side walls **213**. The second insulating member **27** is used for separating the electrode assembly **22** and the side walls **213**, indicating that the second insulating member **27** is disposed between the electrode assembly **22** and the side walls **213** to avoid contact between the electrode assembly **22** and the side walls **213** of the shell **21**, so as to prevent a short circuit in the battery cell **20**.

(79) The second insulating member **27** separates the electrode assembly **22** and the side walls **213** to prevent the contact between the electrode assembly **22** and the side walls **213** from causing a short circuit in the battery cell **20**, so as to improve the safety of the battery cell **20**.

(80) As shown in FIG. **10** and FIG. **11**, in some embodiments, the second insulating member **27** is integrally formed with the first insulating member **26**.

(81) In other embodiments, the second insulating member **27** may be split from the first insulating

member **26**, or the first insulating member **26** and the second insulating member **27** that are split may be connected into a whole.

(82) The second insulating member **27** is integrally formed with the first insulating member **26**, which can better separate the bottom wall **212** of the shell **21** and the electrode assembly **22**, as well as the side walls **213** of the shell **21** and the electrode assembly **22**, to avoid a short circuit in the battery cell **20**, so as to improve the safety performance of the battery cell **20**.

(83) With reference to FIG. **11** and FIG. **12**, in some embodiments, the side walls **213** include two first side walls **2131** opposite in a fourth direction M and two second side walls **2132** opposite in a fifth direction N; the second insulating member **27** includes two first separation portions **271** and second separation portions **272**, the two first separation portions **271** are connected to two ends of the first insulating member **26** in the fourth direction M, two ends of each first separation portion **271** in the fifth direction N are connected with the second separation portions **272**, the two first separation portions **271** are used for separating the two first side walls **2131** and the electrode assembly **22**, the second separation portions **272** are used for separating the second side walls **2132** and the electrode assembly **22**, and the fourth direction M, the fifth direction N, and the thickness direction X of the bottom wall are perpendicular to each other. The first separation portions **271** are connected to the first insulating member **26** in a foldable manner, and the first separation portions **271** are folded around first creases **262** relative to the first insulating member **26**, so that the first separation portions **271** and the first insulating member **26** can be in a vertical state. The second separation portions **272** are connected to the first separation portions **271** in a foldable manner, and the second separation portions **272** are folded around second creases **273** relative to the first separation portions **271**, so that the first separation portions **271** and the second separation portions **272** can be in a vertical state. The second separation portions **272** at the same end of the two first separation portions **271** in the fifth direction N can or cannot overlap after folded relative to the first separation portions **271**, so as to jointly separate the second side walls **2132** and the electrode assembly **22**.

(84) The fifth direction N may be the same as or different from the arrangement direction of the two bent portions II of the electrode assembly **22**. In an embodiment where the fifth direction N is the same as the arrangement direction of the two bent portions II of the electrode assembly **22**, the fourth direction M is the same as the arrangement direction of the plurality of electrode assemblies **22**.

(85) In an embodiment where the battery cell **20** includes a plurality of electrode assemblies **22** arranged side by side, the two first separation portions **271** are located at two ends of an overall structure formed by the plurality of electrode assemblies **22** in the arrangement direction.

(86) As shown in FIG. **13**, in some embodiments, the battery cell **20** further includes a support member **28**, the support member **28** is disposed between the first insulating member **26** and the bottom wall **212**, and the support member **28** is provided with a second through hole **281** corresponding to the first through hole **261**.

(87) The support member **28** may be split from the first insulating member **26**, or the support member **28** and the first insulating member **26** may be split and then connected into an overall structure. Alternatively, the support member **28** and the first insulating member **26** may be integrally formed.

(88) The support member **28** may be positioned with the assembly apparatus **2200** for assembling the battery cell **20** via the second through hole **281**, thereby improving the assembly quality. The support member **28** is provided with the second through hole **281** corresponding to the first through hole **261**, which may be understood as the first through hole **261** and the second through hole **281** are disposed coaxially to facilitate accurate positioning between the first insulating member **26** and the support member **28**, so as to improve the assembly quality.

(89) The support member **28** is disposed between the first insulating member **26** and the bottom wall **212**, which can raise the position of the electrode assembly **22** relative to the bottom wall **212**,

and reduce the risk of wrinkling of the electrode plates due to interference between the connection positions of the side walls **213** and bottom wall **212** of the shell **21** and the electrode plates, thereby reducing the risk of a short circuit in the battery cell **20**.

(90) An embodiment of the present application further provides a battery **100**. The battery **100** includes the battery cell **20** provided by any of the foregoing embodiments.

(91) There is no channel extending in the thickness direction X of the bottom wall between the electrode assembly **22** of the battery cell **20** and the bottom wall **212** of the shell **21**, thereby eliminating a linear ion channel between the electrode assembly **22** and the bottom wall **212**, reducing the risk that ions of the electrode assembly **22** can reach the shell **21** via the first through hole **261**, reducing the risk that the ions of the electrode assembly **22** reach the shell **21** to cause a short circuit in the battery cell **20**, and improving the safety performance of the battery **100** accordingly.

(92) An embodiment of the present application further provides an electrical device, which includes the battery cell **20** provided by any of the foregoing embodiments.

(93) The battery cell **20** provides electrical energy for the electrical device. There is no channel extending in the thickness direction X of the bottom wall between the electrode assembly **22** of the battery cell **20** and the bottom wall **212** of the shell **21**, thereby eliminating a linear ion channel between the electrode assembly **22** and the bottom wall **212**, reducing the risk that ions of the electrode assembly **22** can reach the shell **21** via the first through hole **261**, reducing the risk that the ions of the electrode assembly **22** reach the shell **21** to cause a short circuit in the battery cell **20**, and improving the power safety of the electrical device accordingly.

(94) As shown in FIG. **14**, an embodiment of the present application further provides a manufacturing device **2000** for a battery cell. The manufacturing device **2000** for a battery cell includes a provision apparatus **2100** and an assembly apparatus **2200**; the provision apparatus **2100** is configured to provide a shell **21**, an electrode assembly **22**, and a first insulating member **26**, where the shell **21** has an opening **211** and a bottom wall **212** opposite to the opening **211**; the assembly apparatus **2200** is configured to accommodate the first insulating member **26** and the electrode assembly **22** in the shell **21**, so that the first insulating member **26** separates the electrode assembly **22** and the bottom wall **212**; and the first insulating member **26** is provided with a first through hole **261**, and projections of the electrode assembly **22** and the first through hole **261** on the bottom wall **212** do not overlap in a thickness direction X of the bottom wall.

(95) As shown in FIG. **15**, an embodiment of the present application further provides a manufacturing method for a battery cell **20**. The manufacturing method for the battery cell **20** includes: **S100**: Provide a shell **21**, an electrode assembly **22**, and a first insulating member **26**, where the shell **21** has an opening **211** and a bottom wall **212** opposite to the opening **211**; and **S200**: Accommodate the first insulating member **26** and the electrode assembly **22** in the shell **21**, so that the first insulating member **26** separates the electrode assembly **22** and the bottom wall **212**.

(96) The first insulating member **26** is provided with a first through hole **261**, and projections of the electrode assembly **22** and the first through hole **261** on the bottom wall **212** do not overlap in a thickness direction X of the bottom wall.

(97) An embodiment of the present application provides a battery cell **20**. The battery cell **20** includes a shell **21**, an electrode assembly **22**, a first insulating member **26**, and a second insulating member **27**. The shell **21** includes a bottom wall **212** and side walls **213**, and the side walls **213** are disposed on a periphery of the bottom wall **212** and connected to the bottom wall **212**; the electrode assembly **22** is accommodated in the shell **21**; and the first insulating member **26** is used for separating the electrode assembly **22** and the bottom wall **212**, and the second insulating member **27** is used for separating the electrode assembly **22** and the side walls **213**. The first insulating member **26** and the second insulating member **27** are integrally formed. The first insulating member **26** is provided with a first through hole **261**, and projections of the electrode assembly **22** and the first through hole **261** on the bottom wall **212** do not overlap in a thickness direction of the

bottom wall **212**. A linear ion channel between the electrode assembly **22** and the bottom wall **212** is eliminated, the risk that ions of the electrode assembly **22** can reach the shell **21** via the first through hole **261** is reduced, and the risk that the ions of the electrode assembly **22** reach the shell **21** to cause a short circuit in the battery cell **20** is also reduced. Moreover, as the risk that the ions of the electrode assembly **22** reach the shell **21** is reduced, the risk of corrosion of the shell **21** after the ions are in contact with the shell **21** can also be reduced.

(98) Described above are merely some embodiments of the present application, and the present application is not limited thereto. Various modifications and variations may be made to the present application for those skilled in the art. Any modifications, equivalent substitutions, improvements, and the like made within the spirit and principle of the present application shall fall within the protection scope of the present application.

Claims

1. A battery cell, comprising: a shell having an opening and a bottom wall opposite to the opening; an electrode assembly accommodated in the shell; a first insulating member used for separating the electrode assembly and the bottom wall; and a support member, wherein the support member is disposed between the first insulating member and the bottom wall; wherein: the first insulating member is provided with a first through hole, and projections of the electrode assembly and the first through hole on the bottom wall do not overlap in a thickness direction of the bottom wall; and the support member is provided with a second through hole corresponding to the first through hole, and the first through hole and the second through hole are disposed coaxially.
2. The battery cell according to claim 1, wherein the electrode assembly has a first end face facing the bottom wall, and the first end face exposes electrode plates of the electrode assembly.
3. The battery cell according to claim 2, wherein: the electrode assembly comprises a first electrode plate and a second electrode plate with opposite polarities, and the first electrode plate and the second electrode plate are wound around a winding axis perpendicular to the bottom wall; the electrode assembly comprises a separator, a plurality of first electrode plates, and a plurality of second electrode plates, wherein the first electrode plates and the second electrode plates have opposite polarities, the plurality of first electrode plates and the plurality of second electrode plates are alternately laminated in a first direction, the separator is continuously bent and comprises a plurality of first laminated segments and a plurality of first bent segments, each of the first bent segments is used for connecting two adjacent first laminated segments, the plurality of first bent segments are distributed at two ends of the electrode assembly in a second direction, and the first direction, the second direction, and the thickness direction of the bottom wall are perpendicular to each other; or the electrode assembly comprises a plurality of first electrode plates and a second electrode plate, the first electrode plates and the second electrode plate have opposite polarities, the second electrode plate is continuously bent and comprises a plurality of second laminated segments and a plurality of second bent segments, the plurality of second laminated segments and the plurality of first electrode plates are alternately laminated in the first direction, each of the second bent segments is used for connecting two adjacent second laminated segments, the plurality of second bent segments are distributed at two ends of the electrode assembly in the second direction, and the first direction, the second direction, and the thickness direction of the bottom wall are perpendicular to each other.
4. The battery cell according to claim 2, wherein: the shell has side walls, and the side walls are disposed on a periphery of the bottom wall; and the electrode assembly is a wound electrode assembly and comprises a straight portion and two bent portions, the two bent portions are connected to two ends of the straight portion separately, and the first through hole is disposed between the bent portions and the side walls.
5. The battery cell according to claim 2, wherein the shell further includes side walls, and the side

walls are disposed on a periphery of the bottom wall; the battery cell further comprising: a second insulating member, wherein the second insulating member is used for separating the electrode assembly and the side walls.

6. The battery cell according to claim 1, wherein: the electrode assembly comprises a first electrode plate and a second electrode plate with opposite polarities, and the first electrode plate and the second electrode plate are wound around a winding axis perpendicular to the bottom wall; or the electrode assembly comprises a separator, a plurality of first electrode plates, and a plurality of second electrode plates, wherein the first electrode plates and the second electrode plates have opposite polarities, the plurality of first electrode plates and the plurality of second electrode plates are alternately laminated in a first direction, the separator is continuously bent and comprises a plurality of first laminated segments and a plurality of first bent segments, each of the first bent segments is used for connecting two adjacent first laminated segments, the plurality of first bent segments are distributed at two ends of the electrode assembly in a second direction, and the first direction, the second direction, and the thickness direction of the bottom wall are perpendicular to each other; or the electrode assembly comprises a plurality of first electrode plates and a second electrode plate, the first electrode plates and the second electrode plate have opposite polarities, the second electrode plate is continuously bent and comprises a plurality of second laminated segments and a plurality of second bent segments, the plurality of second laminated segments and the plurality of first electrode plates are alternately laminated in the first direction, each of the second bent segments is used for connecting two adjacent second laminated segments, the plurality of second bent segments are distributed at two ends of the electrode assembly in the second direction, and the first direction, the second direction, and the thickness direction of the bottom wall are perpendicular to each other.

7. The battery cell according to claim 6, wherein the shell further includes side walls, and the side walls are disposed on a periphery of the bottom wall; the battery cell further comprising: a second insulating member, wherein the second insulating member is used for separating the electrode assembly and the side walls.

8. The battery cell according to claim 1, wherein: the shell has side walls, and the side walls are disposed on a periphery of the bottom wall; and the electrode assembly is a wound electrode assembly and comprises a straight portion and two bent portions, the two bent portions are connected to two ends of the straight portion separately, and the first through hole is disposed between the bent portions and the side walls.

9. The battery cell according to claim 8, wherein the first insulating member is rectangular, and the first through hole is disposed in at least one of four corners of the first insulating member.

10. The battery cell according to claim 9, wherein: the battery cell comprises a plurality of electrode assemblies arranged side by side; and in an arrangement direction of the plurality of electrode assemblies, the first through hole is disposed between two adjacent bent portions.

11. The battery cell according to claim 9, wherein two groups of first through holes are disposed between two adjacent electrode assemblies, the two groups of first through holes are spaced, and some straight portions are located between the two groups of first through holes.

12. The battery cell according to claim 8, wherein: the battery cell comprises a plurality of electrode assemblies arranged side by side; and in an arrangement direction of the plurality of electrode assemblies, the first through hole is disposed between two adjacent bent portions.

13. The battery cell according to claim 12, wherein two groups of first through holes are disposed between two adjacent electrode assemblies, the two groups of first through holes are spaced, and some straight portions are located between the two groups of first through holes.

14. The battery cell according to claim 8, wherein two groups of first through holes are disposed between two adjacent electrode assemblies, the two groups of first through holes are spaced, and some straight portions are located between the two groups of first through holes.

15. The battery cell according to claim 1, wherein the shell further includes side walls, and the side

walls are disposed on a periphery of the bottom wall; the battery cell further comprising: a second insulating member, wherein the second insulating member is used for separating the electrode assembly and the side walls.

16. The battery cell according to claim 15, wherein the second insulating member is integrally formed with the first insulating member.

17. A battery, comprising the battery cell according to claim 1.

18. An electrical device, comprising the battery cell according to claim 1.
